

HYDROGENATED NBR (HNBR): PATENT LANDSCAPE ON HYDROGENATION, LOW-MOONEY GRADES, AND CATALYST-RESIDUE CONTROL

1. ABSTRACT

This report synthesizes six primary patents relevant to hydrogenated nitrile rubber (HNBR), focusing on how NBR precursors are transformed into hydrogenated derivatives with reduced residual unsaturation, improved heat/oil/ozone resistance, and better processability. The landscape splits into two major technical directions: (1) direct hydrogenation of NBR using noble-metal catalysts under controlled pressure, temperature, and catalyst loading, and (2) simultaneous metathesis/hydrogenation routes designed to lower molecular weight and Mooney viscosity in a single step. A third theme is catalyst-residue management, where the objective is not only to hydrogenate efficiently but also to minimize residual Ru/Rh/Pd that can impair ageing, discoloration, and gel stability.

US20210340285A1 and EP3827029A1 disclose a solution hydrogenation process for NBR using ruthenium, palladium, or rhodium compounds, with explicit operating windows of 60 to 200°C, 700,000 to 15,000,000 Pa, and 1 to 24 hours. These patents emphasize that catalyst amount and post-reaction residue control are central to achieving HNBR with improved ageing properties. US7951875B2 and EP2473281B1 instead target low-molecular-weight HNBR via metathesis-assisted degradation/hydrogenation, using ruthenium-based catalyst systems and co-olefins such as ethylene or 1-hexene; the disclosed examples show substantial reductions in Mw, Mn, PDI, and Mooney viscosity. US8957161B2 is related more indirectly: it teaches salt-accelerated metathesis of nitrile rubber using Ru catalysts, which is relevant as a precursor degradation strategy for HNBR feedstocks. US9150669B2 further develops the simultaneous metathesis/hydrogenation concept and demonstrates that catalyst choice strongly affects hydrogenation rate and final polymer properties.

Across the patents, the dominant process trends are solution-phase operation, ruthenium-centered catalyst chemistry, use of co-olefins to drive metathesis, and explicit attention to residual double bond content and catalyst residues. The evidence supports a synthesis strategy combining controlled hydrogenation conditions with catalyst systems that either preserve polymer integrity or intentionally reduce molecular weight before/while hydrogenating.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis of nitrile rubber in the presence of a catalyst system comprising a metathesis catalyst and one or more salts, with the salts accelerating the metathesis reaction. The process is presented as a way to degrade nitrile rubber prior to or in connection with HNBR-related processing, rather than as a direct hydrogenation patent. - Salt: LiBr — Example 7 - Salt amount: 1 mg LiBr — Example 7.02 - Salt: CsBr — Example 8 - Salt amount: 1 mg CsBr — Example 8.02 - Metathesis catalyst: Grubbs II catalyst — Example 7/8 - Metathesis catalyst amount: 10 mg — Example 7/8 - Solvent: chlorobenzene — Example 7 - Solvent amount: 0.3 ml chlorobenzene — Example 7 - Solvent: CDCl₃ — Example 7 - Solvent amount: 0.2 ml CDCl₃ — Example 7 - Reaction temperature: room temperature — Example 7/8 - Substrate amount: 0.15 ml diethyl diallylmalonate — Example 7 - Time: 30 min — Example 7.01/7.02 - Time: 60 min — Example 7.01/7.02 - Time: 15 min — Example 8.01/8.02 - Time: 30 min — Example 8.01/8.02 - Time: 60 min — Example 8.01/8.02 - Time: 90 min — Example 8.01/8.02 - Time: 150 min — Example 8.01/8.02 - Weight ratio of salt(s): metathesis catalysts: 0.01:1 to 10000:1 — Claim 1

Relevant Experimental Evidence - Example 7: LiBr addition accelerated ring-closing metathesis relative to the salt-free control. - Example 8: CsBr addition was also tested and showed altered conversion behavior versus the control.

Technical Relevance Although not a direct hydrogenation patent, it is relevant because it teaches salt-accelerated Ru-catalyzed metathesis of nitrile rubber, a degradation strategy used to tune NBR feedstock before HNBR formation. The disclosed catalyst/salt system informs molecular-weight control routes that can precede hydrogenation.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by a metathesis-based degradation of nitrile rubber followed by hydrogenation, using ruthenium-based catalyst systems and co-olefins. The process is designed to reduce molecular weight and narrow molecular weight distribution while achieving high hydrogenation levels. - Rubber concentration: 15 wt.-% solid — Example 1 - Rubber concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - 1-Hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity: 73 — Substrate - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4 - Mn: 85 kg/mol — Substrate - Mw: 296 kg/mol — Substrate - PDI: 3.4 — Substrate - Mn: 27 kg/mol — Typical product - Mw: 54 kg/mol — Typical product - PDI: 2.0 — Typical product - Mw: 189000 — Experiment 1 - Mn: 73000 — Experiment 1 - PDI: 2.59 — Experiment 1 - Mw: 136000 — Experiment 2 - Mn: 60000 — Experiment 2 - PDI: 2.27 — Experiment 2 - Mw: 59000 — Experiment 3 - Mn: 31000 — Experiment 3 - PDI: 1.90 — Experiment 3 - Mw: 111000 — Experiment 4 - Mn: 55000 — Experiment 4 - PDI: 2.02 — Experiment 4 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 100% — Table 2 - RDB: 2.5% — Table 2 - RDB: 0.6% — Table 2 - RDB: 0.5% — Table 2 - RDB: 0.4% — Table 2 - RDB: 7.0% — Table 2 - RDB: 4.5% — Table 2 - RDB: 3.9% — Table 2 - RDB: 3.5% — Table 2 - RDB: 15.4% — Table 2 - RDB: 3.1% — Table 2 - RDB: 1.3% — Table 2 - RDB: 0.7% — Table 2 - Hydrogenation catalyst amount: 0.041 phr — Table 1 - Hydrogenation catalyst amount: 0.045 phr — Table 1 - Hydrogenation catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr triphenylphosphine — Table 1

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave rapid hydrogenation and reduced Mw/Mn relative to the substrate. - Example 2: Arlt catalyst also produced low-RDB HNBR with reduced molecular weight. - Example 3: Lower solid content and ethylene feed produced the lowest Mooney viscosity and narrowest PDI. - Example 4: 1-hexene co-olefin was used and still yielded a lower-Mw HNBR product. - Example 1: The catalyst comparison showed HNBR 1 outperforming the industrial norm in hydrogenation efficiency.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR, a key need for improved processability. It provides concrete operating and formulation data linking catalyst choice, co-olefin, and hydrogenation conditions to reduced Mooney viscosity and narrower molecular-weight distribution.

Patent 3

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of at least partially unsaturated nitrile rubber using ruthenium, palladium, or rhodium compounds, with an emphasis on producing HNBR that retains good ageing properties without a separate catalyst-removal step. The process is defined by catalyst loading, temperature, pressure, and reaction time windows. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200° C. — Claim 1 - Hydrogenation pressure: 700000 to 15000000 Pa — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Ageing test temperature: 140° C. — Description - Ageing test duration: 4 days — Description - Gel content increase: less than 10% — Description - Mooney viscosity increase: less than 40% — Description - HNBR composition: 40% to 90% by weight conjugated diene units — Claim 1 - HNBR composition: 10% to 60% by weight α,β -unsaturated nitrile units — Claim 1

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it defines practical hydrogenation windows for making HNBR from NBR in solution while controlling catalyst residue and ageing performance. It is especially important for industrial synthesis because it balances catalyst activity, reaction severity, and downstream product stability.

Patent 4

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This European counterpart to US20210340285A1 discloses solution hydrogenation of at least partially unsaturated nitrile rubber using Ru, Pd, or Rh compounds to produce HNBR with improved ageing properties. The patent emphasizes catalyst amount control and the avoidance of elaborate post-hydrogenation metal-removal steps. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200° C. — Claim 1 - Hydrogenation pressure: 700000 to 15000000 Pa — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Ageing test temperature: 140° C. — Description - Ageing test duration: 4 days — Description - Gel content increase: less than 10% — Description - Mooney viscosity increase: less than 40% — Description - HNBR composition: 40% to 90% by weight conjugated diene units — Claim 1 - HNBR composition: 10% to 60% by weight α,β -unsaturated nitrile units — Claim 1

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it provides the same core HNBR hydrogenation window as the US counterpart, with explicit catalyst loading and operating conditions. It is useful for identifying industrially viable synthesis parameters that preserve ageing properties and reduce catalyst-residue issues.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneously subjecting nitrile rubber to metathesis and hydrogenation in the presence of hydrogen and a ruthenium/osmium catalyst system. The process is intended to lower molecular weight and Mooney viscosity while achieving low residual double bond content. - Molecular weight Mw: 20,000 to 250,000 — Description - Mooney viscosity: 1 to 50 — Description - MWD/PDI: less than 3.0 — Description - Mooney

viscosity of nitrile rubber feed: 24 to 70 — Description - Preferred Mooney viscosity of nitrile rubber feed: 28 to 40 — Description - Ethylene pressure: 6.9×10^5 Pa — Example context in cited prior art discussion - Conjugated diene: C4-C6 — Description - Preferred conjugated dienes: 1,3-butadiene, isoprene, 2,3-dimethylbutadiene, piperylene — Description - Particular preference: 1,3-butadiene and isoprene — Description - Especial preference: 1,3-butadiene — Description - α,β -unsaturated nitrile: C3-C5 — Description - Preferred nitriles: acrylonitrile, methacrylonitrile, ethacrylonitrile — Description - Further comonomers: fumaric acid, maleic acid, acrylic acid, methacrylic acid — Description - Hydrogenation/metathesis in presence of hydrogen: yes — Claim 1 - Co-olefin: optionally at least one co-olefin — Claim 1

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalyst systems were compared at equal metal content. - Example 1: HNBR 1 reached RDB 0.4% after 240 min and had Mw 113371 g/mol, Mn 48352 g/mol, PDI 2.34, and Mooney 33.3. - Example 1: HNBR 2 gave Mw 133758 g/mol, Mn 54595 g/mol, PDI 2.45, and Mooney 41.3. - Example 1: HNBR 3 gave the highest Mw and Mooney viscosity, showing the least favorable performance among the compared catalysts.

Technical Relevance This patent is central to HNBR synthesis because it combines molecular-weight reduction and hydrogenation in a single process, directly addressing the low-Mooney, low-RDB target. The example data show how catalyst selection affects hydrogenation efficiency and final polymer rheology.

Patent 6

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a single-step process for producing low-molecular-weight HNBR by contacting nitrile rubber with a ruthenium or osmium catalyst in the presence of hydrogen, optionally with a co-olefin. The process is explicitly framed as simultaneous metathesis and hydrogenation, yielding HNBR with lower molecular weight and narrower molecular-weight distribution. - Molecular weight Mw: 20,000 to 250,000 — Description - Mooney viscosity: 1 to 50 — Description - MWD/PDI: less than 3.0 — Description - Conjugated diene: C4-C6 — Description - Preferred conjugated dienes: 1,3-butadiene, isoprene, 2,3-dimethylbutadiene, piperylene — Description - Particular preference: 1,3-butadiene and isoprene — Description - Especial preference: 1,3-butadiene — Description - α,β -unsaturated nitrile: C3-C5 — Description - Preferred nitriles: acrylonitrile, methacrylonitrile, ethacrylonitrile — Description - Further comonomers: fumaric acid, maleic acid, acrylic acid, methacrylic acid —

Description - Hydrogen in reaction medium: present — Claim 1 - Co-olefin: optionally at least one co-olefin — Claim 1 - Catalyst family: ruthenium or osmium compounds — Claim 1 - Catalyst comparison metal content: 65.6 ppm — Example 1 - Hydrogenation catalyst amount: 0.041 phr — Example 1 - Hydrogenation catalyst amount: 0.045 phr — Example 1 - Hydrogenation catalyst amount: 0.060 phr — Example 1 - Co-catalyst: triphenylphosphine — Example 1 - Co-catalyst amount: 1.0 phr — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% — Example 1 - RDB: 2.5% — Example 1 - RDB: 0.6% — Example 1 - RDB: 0.5% — Example 1 - RDB: 0.4% — Example 1 - RDB: 7.0% — Example 1 - RDB: 4.5% — Example 1 - RDB: 3.9% — Example 1 - RDB: 3.5% — Example 1 - RDB: 15.4% — Example 1 - RDB: 3.1% — Example 1 - RDB: 1.3% — Example 1 - RDB: 0.7% — Example 1 - Mw: 113371 g/mol — Example 1 - Mn: 48352 g/mol — Example 1 - PDI: 2.34 — Example 1 - Mooney viscosity: 33.3 — Example 1 - Mw: 133758 g/mol — Example 1 - Mn: 54595 g/mol — Example 1 - PDI: 2.45 — Example 1 - Mooney viscosity: 41.3 — Example 1 - Mw: 195304 g/mol — Example 1 - Mn: 64130 g/mol — Example 1 - PDI: 3.05 — Example 1 - Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared at equal metal content of 65.6 ppm. - Example 1: HNBR 1 achieved the fastest hydrogenation, reaching RDB 0.4% after 240 min. - Example 1: HNBR 1 gave the lowest Mooney viscosity, 33.3, and the lowest Mw among the three catalysts. - Example 1: HNBR 3, the industrial norm, gave the highest Mw and Mooney viscosity, showing the least favorable low-MW performance.

Technical Relevance This patent is highly relevant because it directly teaches the simultaneous metathesis/hydrogenation route for producing low-Mooney HNBR in one step. The example data provide a practical benchmark for catalyst selection and show how catalyst system choice controls residual unsaturation and molecular weight.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US8957161B2 focuses on salt-accelerated Ru-catalyzed metathesis of nitrile rubber, whereas US7951875B2, EP2473281B1, and US9150669B2 use metathesis/hydrogenation to directly generate low-MW HNBR.
- US20210340285A1 and EP3827029A1 are the only patents in the set that explicitly define direct hydrogenation windows for NBR to HNBR: 60 to 200°C, 700,000 to 15,000,000 Pa, and 1 to 24 hours, with Ru/Pd/Rh catalyst loadings in the ppm range.
- US7951875B2 and US9150669B2 both disclose catalyst screening at 65.6 ppm metal content and show that catalyst identity strongly affects RDB decay, Mw, Mn, PDI, and Mooney viscosity after 240 min.

- EP2473281B1 and US9150669B2 both claim low-MW HNBR with Mw 20,000 to 250,000, Mooney 1 to 50, and PDI below 3.0; EP2473281B1 reports a preferred feed Mooney of 28 to 40, while US9150669B2 emphasizes the same product window but with a single-step simultaneous process.
- The metathesis-based patents rely on ruthenium/osmium catalyst families and optionally co-olefins; US7951875B2 explicitly uses ethylene at 6.9×10^5 Pa and 1-hexene, while the other metathesis patents describe co-olefin use more generally.
- Catalyst-residue and ageing concerns are central in US20210340285A1 and EP3827029A1, which link noble-metal content to discoloration and ageing performance, whereas the metathesis patents prioritize molecular-weight reduction and residual double bond control.
- Across the set, the most consistent process trend is solution-phase processing with ruthenium-centered chemistry; the most consistent product trend is lowering residual unsaturation while reducing Mooney viscosity and narrowing molecular-weight distribution.

4. CONCLUSION

The evidence supports two complementary synthesis strategies for HNBR: direct solution hydrogenation of NBR with controlled Ru/Pd/Rh loading and temperature/pressure windows, and simultaneous metathesis-hydrogenation to produce low-MW, low-Mooney HNBR in a single step. For industrially robust HNBR synthesis, the strongest evidence favors solution hydrogenation at 60 to 200°C and 700,000 to 15,000,000 Pa with ppm-level noble-metal catalysts when ageing stability and catalyst-residue control are priorities. When low viscosity and improved processability are the main targets, the metathesis/hydrogenation patents indicate that ruthenium-based catalyst systems, optionally with co-olefins such as ethylene or 1-hexene, can reduce Mw into the 20,000 to 250,000 range and Mooney viscosity into the 1 to 50 range while maintaining low residual double bond content. The most defensible recommended synthesis parameters from the evidence are: solution-phase operation, Ru-centered catalyst systems in the tens of ppm range, hydrogenation at 60 to 200°C under 0.7 to 15 MPa, and, for low-MW grades, simultaneous metathesis/hydrogenation with co-olefin assistance.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>